

9.8 Introduction to Acoustics

The branch of Physics that deals with the process of generation, reception and propagation of sound is called "Acoustics". Thus acoustics is the science of sound. Originally it was limited to stimulation of the human ear by the sound incident. Modern acoustics deals with all sorts of sounds which have no relation to the human ear. Examples are seismological disturbances and ultrasonics. Modern acoustics as a science and technology has experienced enormous development. The practical applications are multifarious. They include:

Architectural acoustics – study of sound waves in closed rooms

- Underwater acoustics – use of sound in sea for the detection of vessels and the exploration of sea bed.
- Engineering acoustics – technology of sound production and recording, study of vibrations of solids and their control as well as noise control
- Musical acoustics – the physics of musical instruments
- Bio-acoustics/medical acoustics – use of sound in medical diagnosis and therapy

Basic acoustics may be divided into three branches namely, production, transmission and detection of sound. First let us see some basic concepts of sound.

9.9 Classification of Sound

Sound is a mechanical wave. It is propagated through a medium such as air by means of wave motion. The sound waves may be classified based on their frequency (Fig.33) as follows:

- i) infrasonic – frequency less than 20 Hz (inaudible)
- ii) sonic – frequency having range between 20 Hz and 20 kHz (audible)
- iii) ultrasonic – frequency more than 20 kHz (inaudible)

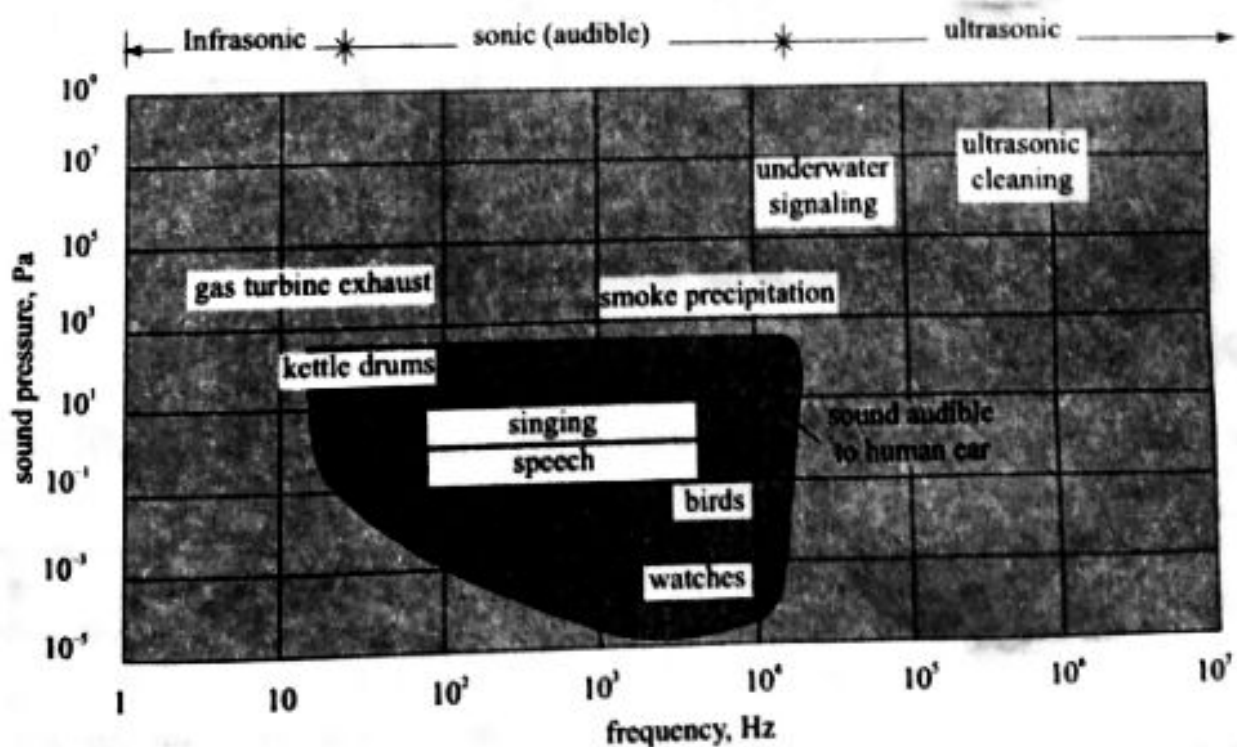


Fig. 33 Spectrum of sound

Sonic (audible) sound may be further classified as

- i) musical sound and
- ii) 'noise'

9.9.1 Musical sound

The sound that produces a pleasing effect or sensation on the ear is called a musical sound. Musical sounds are characterised by the following properties of time Vs intensity curves called **sound waveforms**.

- a) regularity in shape of the curve
- b) definite periodicity in the curves and
- c) absence of any sudden changes in their intensities

Fig.34(a) represents the waveform of few musical sounds while Fig.34(b) represents the waveform of a noise.

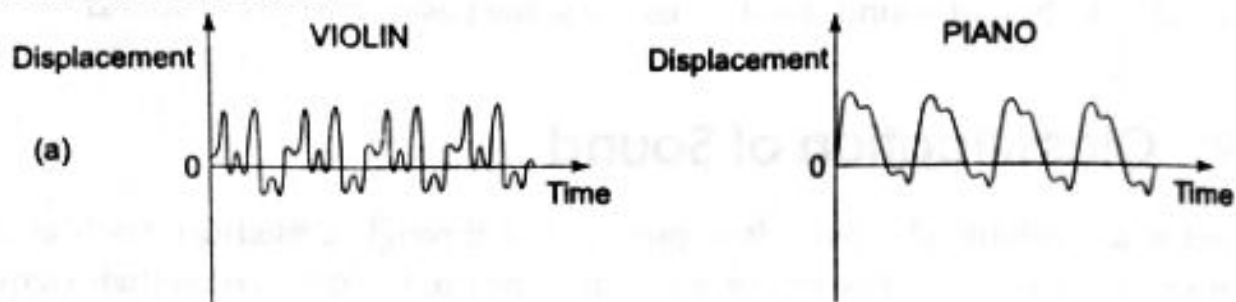


Fig.34 (a) Waveform of musical sound

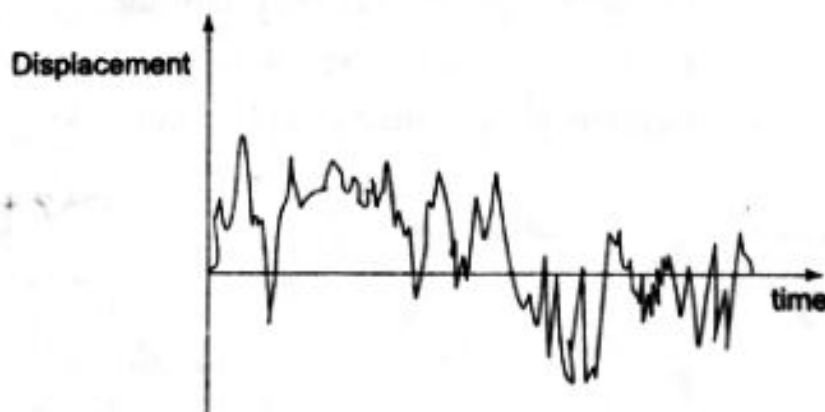


Fig.34 (b) Waveform of a noise

9.9.2 Noise

The sound that produces a displeasing (irritating) effect or sensation on the ear is called a noise. The waveforms produced by noise

- a) are irregular in shape
- b) lack in periodicity and
- c) undergo sudden changes in their intensities.

9.10 Characteristics of Musical Sounds

There are three characteristics of musical sounds:

- i) Pitch or frequency
- ii) Loudness or intensity and
- iii) Quality or timbre

9.10.1 Pitch or frequency

The pitch of a sound is a physiological quantity felt due to mental sensation experienced by the observer and it depends upon the frequency. It does not depend upon loudness or quality. It is directly proportional to the frequency.

Although pitch of a sound is directly proportional to its frequency, they are different. Frequency of a sound is defined as the number of vibrations per second. This physical quantity can be measured accurately, while the pitch of a sound is a physiological quantity which cannot be measured.

A high frequency source produces a shrill sound while a low frequency source produces lower pitch. The voice produced by ladies and children has high pitch because the frequency is high when compared to gents voice. The sound produced by a mosquito is of high pitch due to high frequency while the sound produced by a roaring lion is of low pitch due to low frequency.

You may be interested to note that the frequencies of eight standard tuning forks correspond to the pitch of the eight musical notes. While the frequency of the first tuning fork 256 Hz corresponds to the note 'Sa', the frequency of the eighth tuning fork 512 Hz (which is the overtone of the first i.e. 256 Hz) corresponds to the note 'Sa'. In Table 1, the frequencies of eight standard tuning forks, their corresponding pitch and the set of letters used to denote them are given.

Table 1 Pitch of the note produced by standard tuning forks

Freq. in Hz	256	288	320	341.3	384	426.7	480	512
Indian swara notation	(Sa)	(ri)	(ga)	(ma)	(pa)	(dha)	(ni)	(sa')
symbol	C	D	E	F	G	A	B	C'

9.10.2 Loudness or intensity

A source while creating a sound pushes and pulls the air doing work on it at some rate. The air is made to vibrate; then the vibration and hence the energy travel away from the source (Fig.35). With increase of distance from the source, the intensity of the sound wave decreases.

The intensity of the sound wave is defined as the rate of flow of sound energy through a unit area chosen perpendicular to the direction of propagation. Since the rate of flow of energy is power, the intensity of sound wave is measured in units of power per unit area i.e. in watts per square metre. Sound intensity detectable by the ear may range from a low value of about 10^{-12} watt/m² up to about 1 watt/m². To simplify the description of sound intensities, a system

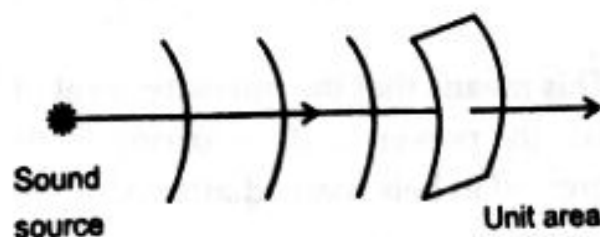


Fig. 35 Flow of sound energy

dealing with exponents of 10 has been devised. This is what is called the **decibel system**. Sounds of same intensity but different frequencies may differ in loudness.

The energy flowing per unit time through unit cross sectional area, represents the intensity of sound

$$I = 2\pi^2 a^2 f^2 \rho v \quad (9)$$

Thus we find that the intensity of sound is directly proportional to the

- i) square of the amplitude of the sound wave
- ii) square of the frequency of the sound wave
- iii) density of the medium and
- iv) velocity of sound in that medium.

9.10.3 Measurement of intensity level-decibel

When the intensity of sound wave is less than 10^{-12} watt/m², human ear cannot sense the sound and hence intensity level corresponding to 10^{-12} watt/m² for sound wave of frequency 1000 Hz is regarded as **zero level of intensity** (I_0) (remember that zero level of intensity does not mean zero intensity).

Though the absolute intensity of sound wave is measured in terms of watt/m², it is the relative intensity which has more practical significance. **The ratio of the intensity of sound wave to the standard zero level of intensity is defined as intensity level of sound.** For example, if $I = 10^{-4}$ watt/m², then

$$\frac{I}{I_0} = \frac{10^{-4} \text{ watt/m}^2}{10^{-12} \text{ watt/m}^2} = 10^8.$$

Human hearing mechanism responds to large range of intensities. Hence a logarithmic scale is convenient. Therefore taking log of the above equation

$$\log \frac{I}{I_0} = \log 10^8 = 8 \log_{10} 10 = 8$$

This means that the intensity level of sound of intensity $I = 10^{-4}$ watt/m² is 8 bels. i.e. the power of 10 occurring in the ratio I/I_0 is the intensity level of sound expressed in bels (named after Alexander Graham Bell).

For convenience, one tenth of a bel is often used as a unit and given a name **decibel**, abbreviated dB, where

$$1 \text{ bel} = 10 \text{ decibels}$$

$$\text{Hence sound intensity level } \beta = 10 \log(I/I_0) \text{ dB} \quad (10)$$

Let us now try to understand what is meant by 1 dB change in intensity level of sound. An increase of one decibel in intensity level of sound corresponds to an increase of 26% of the source intensity (Solved Problem 18). Similarly 2 dB change in

intensity level of sound signifies an increase of $(1.26)^2$ in intensity. This corresponds to an increase of 58% of the intensity of sound. In similar manner we can prove that 3 dB change in intensity level correspond to 100% change of the intensity of sound. (Solved Problem 19). It must be noted that whenever intensity of sound is expressed in dB, it represents intensity level of sound only.

As we have minimum limit for hearing, there also exists a maximum intensity which the human ear can tolerate without sensation of pain. This upper limit of hearing is 1 watt/m^2 which corresponds to 120 dB. This is called **threshold of feeling or pain threshold**. The lower limit of hearing $10^{-12} \text{ watt/m}^2$ called the zero level intensity corresponds to sound level of 0 dB.

9.10.4 Phon

The ear does not have the same sensitivity for sounds of all frequencies. It is most sensitive to low intensity sounds at about 3500 Hz. That is, the intensity at the threshold of hearing is lower at this frequency than at any other. This means that the sounds of same intensity but having different frequencies may differ in loudness. Loudness is not directly measurable with a meter. During measurement, the source whose equivalent loudness has to be measured, is placed near the standard source of 1000 Hz frequency in front of the listener. Two sounds are heard alternatively and the intensity of the standard source is adjusted such that it has the same loudness as the sound of the given source. Then the intensity level of the standard source is measured. If it is N decibels the equivalent loudness is said to be N phons. i.e., N phons means "as loud as N dB at 1000 Hz". Thus the measure of loudness of any sound in phon is equal to the intensity level of an equally loud pure tone of frequency 1000 Hz expressed in decibel.

i.e. for a frequency of 1000 Hz,

$$L_p = 10 \log(I/I_0)$$

where L_p is the loudness level in phons

$$\text{Since } I_0 = 10^{-12} \text{ watt/m}^2$$

$$\begin{aligned} L_p &= 10 \log I + 10 \log \left(\frac{1}{I_0} \right) \\ &= 10 \log I + 10 \log 10^{12} \\ &= (10 \log I + 120) \text{ phon} \end{aligned} \quad (11)$$

where k is a constant. From this law we understand that when the intensity of a sound is doubled the loudness is not doubled. But it can be proved that the loudness increases by the same amount each time the intensity is doubled. Let I_1 be the initial intensity of the source of loudness L_1 . Then

$$L_1 = k \log I_1$$

When the intensity is doubled, I_1 becomes $2I_1$. Hence

$$\begin{aligned} L_2 &= k \log 2I_1 = k(\log 2 + \log I_1) \\ &= k \log 2 + k \log I_1 = k \log 2 + L_1 \end{aligned}$$

Hence increase in loudness $L_2 - L_1 = k \log 2$. Since $k \log 2$ is a constant, everytime intensity is doubled, irrespective of the initial intensity, the loudness increases by the same amount.

9.10.6 Sone

Another scale currently used to measure the loudness is Sone. *One sone is defined as the loudness of a 1000 Hz tone of 40 dB intensity level.* It is also equal to the loudness of any sound having a loudness level of 40 phons.

In the range from 40 phons to 100 phons, empirically one can write the rule of thumb for loudness as

$$\log L_s = 0.03(L_p - 40) \quad (13)$$

where L_s is the loudness in sones and
 L_p is the loudness level in phons

Intensity Phones L_p	$\log L_s$	Loudness Sonos L_s
40	0	1
50	0.3	2
60	0.6	4
70	0.9	8
80	1.2	16
90	1.5	32
100	1.8	64

This rule states that the sound intensity (Phones) must be increased by a factor of 10 to increase the loudness (Sonos) twice.

Solved Problem 16

Find the sound intensity at 3 meters from a speaker that is placed on the side wall of a hall, feeding sound evenly into a hemisphere in front of it. The sound power of the speaker is 2 watts.

$$\text{Sound Intensity} = \frac{\text{Sound power}}{\text{Area spread}}$$

The sound generated by the speaker spreads in the shape of a hemispherical envelope.

$$\text{Area of hemisphere} = 2\pi r^2 = 2\pi \times 3^2 \text{ m}^2 = 56.57 \text{ m}^2$$

Sound intensity is the power per unit area.

$$\text{Hence sound intensity} = \frac{2 \text{ watts}}{56.57 \text{ m}^2} = 3.54 \times 10^{-2} \text{ watt/m}^2$$

$$\text{The sound intensity at 3 metres from the source} = 3.54 \times 10^{-2} \text{ watt/m}^2$$

(Answer)

Solved Problem 17

Find the intensity level in dB of a sound of intensity $3.5 \times 10^{-2} \text{ watt/m}^2$. Zero intensity level = $10^{-12} \text{ watt/m}^2$.

Solution

$$\text{Sound level} = 10 \log(I/I_0)$$

$$\text{Given } I = 3.5 \times 10^{-2} \text{ watts/m}^2$$

$$I_0 = 10^{-12} \text{ watts/m}^2$$

$$\text{Hence sound level} = 10 \log(3.5 \times 10^{-2}/10^{-12})$$

$$= 10 \log(3.5 \times 10^{10})$$

$$= 10 (\log 3.5 + \log 10^{10})$$

$$= 10(0.54 + 10)$$

$$= 105.4 \text{ dB}$$

The sound intensity level will be 105.4 dB

(Answer)

Solved Problem 18

Prove that 1 dB change in intensity level of sound corresponds to an increase of 26% of the intensity of sound.

Solution

$$\text{Sound level } dB_1 = 10 \log \frac{I_1}{I_0}$$

$$dB_2 = 10 \log \frac{I_2}{I_0}$$

$$\text{Hence } dB_2 - dB_1 = 10 \left(\log \frac{I_2}{I_0} - \log \frac{I_1}{I_0} \right)$$

$$\text{i.e. } 1 = 10 \log \frac{I_2}{I_1}$$

$$\text{or } \log \frac{I_2}{I_1} = \frac{1}{10} = 0.1$$

$$\frac{I_2}{I_1} = 10^{0.1} = 1.26$$

$$\text{or } \frac{I_2 - I_1}{I_1} = 0.26 = 26\%$$

(Answer)

Solved Problem 19

If the intensity of a sound in watts/m^2 is doubled, by how much is the dB level changed?

Solution

$$\text{Sound level } dB = 10 \log \frac{I_1}{I_0}$$

Let the first intensity be I_1 and the corresponding intensity level be dB_1 . The intensity is then increased to I_2 where $I_2 = 2I_1$ and the corresponding intensity level be dB_2 . The problem is to find the change $dB_2 - dB_1$.

$$dB_1 = 10 \log \frac{I_1}{I_0}$$

$$dB_2 = 10 \log \frac{2I_1}{I_0}$$

$$dB_2 - dB_1 = 10 \left(\log \frac{I_2}{I_0} - \log \frac{I_1}{I_0} \right)$$

$$= 10 \log \frac{I_2}{I_1}$$

$$= 10 \log \frac{2I_1}{I_1}$$

$$\text{Hence } dB_2 - dB_1 = 10 \log 2$$

$$= 3.01$$

The sound intensity level is increased by 3 dB when the intensity is doubled.

(Answer)

Solved Problem 20

What is the resultant sound level when 70 dB sound is added to 85 dB sound?

Solution

It is the energy flow per unit area i.e. intensity that adds; so to do this problem it is necessary to find the intensity associated with each sound level, add the intensities, and then convert back to intensity level in dB.

Let the intensities be I_1 and I_2 respectively.
For the first sound

$$\begin{aligned} 70 &= 10 \log \frac{I_1}{I_0} \\ \text{i.e. } \log \frac{I_1}{I_0} &= 7 \\ \frac{I_1}{I_0} &= 10^7 \\ \text{Hence } I_1 &= 10^7 I_0 \end{aligned}$$

For the second sound

$$\begin{aligned} 85 &= 10 \log \frac{I_2}{I_0} \\ \log \frac{I_2}{I_0} &= 8.5 \\ \frac{I_2}{I_0} &= 10^{8.5} = 10^{1/2} \times 10^8 = 3.16 \times 10^8 \\ &= 31.6 \times 10^7 \\ I_2 &= 31.6 \times 10^7 I_0 \end{aligned}$$

Adding the two sounds,

$$\begin{aligned} I &= I_1 + I_2 = 10^7 I_0 + 31.6 \times 10^7 I_0 \\ &= 32.6 \times 10^7 I_0 \\ &= 3.26 \times 10^8 I_0 \\ \frac{I}{I_0} &= 3.26 \times 10^8 \end{aligned}$$

For the sum, the intensity level in dB is

$$\begin{aligned} 10 \log (3.26 \times 10^8) &= 10 [\log 3.26 + 8 \log 10] \\ &= 10 [0.513 + 8] = 85.13 \text{ dB} \end{aligned}$$

Hence when 70 dB and 85 dB sounds are added the resultant intensity level will be 85.13 dB. (Answer)

Solved Problem 21

If the intensity of a source of sound is increased to 20 times its value, by how many decibel does the intensity level increase?

Solution

Let the first intensity be I_1 and the corresponding intensity level be dB_1 .

The intensity is then increased to I_2 where $I_2 = 20I_1$ and the corresponding intensity level be dB_2 . Increase in intensity level is $dB_2 - dB_1$.

$$\begin{aligned}
 dB_1 &= 10 \log \frac{I_1}{I_0} \\
 dB_2 &= 10 \log \frac{20I_1}{I_0} \\
 &= 10 \left(\log 20 + \log \frac{I_1}{I_0} \right) \text{ (Since } \log ab = \log a + \log b \text{)} \\
 &= 10 \log 20 + \log \frac{I_1}{I_0} \\
 &= 10 \times 1.30 + dB_1
 \end{aligned}$$

Increase in intensity level $dB_2 - dB_1 = 13 \text{ dB}$

(Answer)

9.10.7 Quality or Timbre

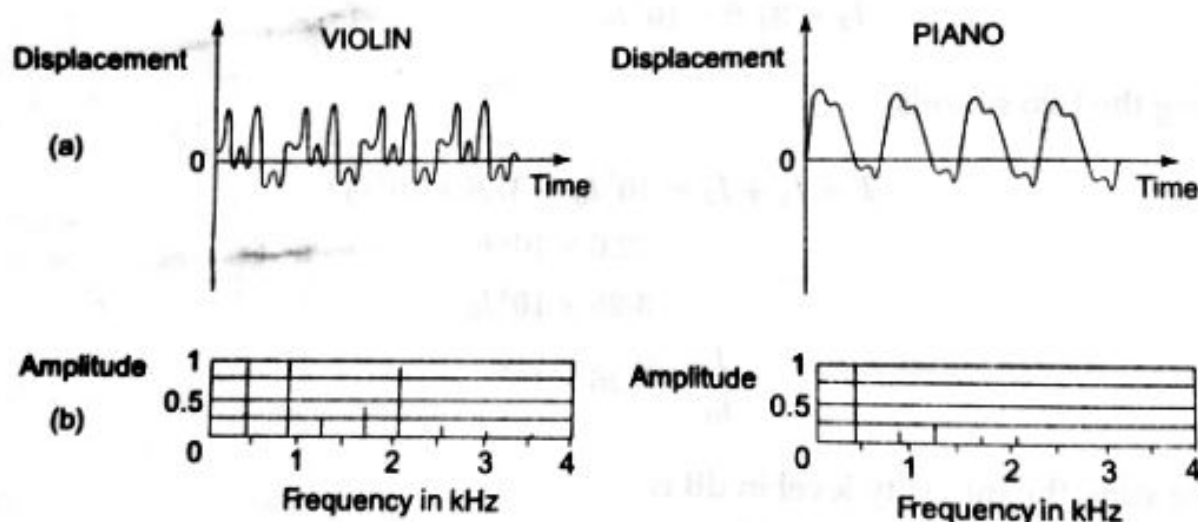


Fig. 36 (a) waveforms of violin and piano and (b) their respective frequency spectra

The third characteristic of a musical note is **quality** or **timbre**. The notes emitted by musical instruments are never simple. They are not composed of a single frequency alone. Normally they consist of a large number of tones of different frequencies of varying intensity. The tone of the smallest frequency is the loudest and is called the **fundamental**. This frequency is dominant and defines the pitch of the note. The frequencies accompanying the fundamental are called **overtones** or upper **partials**. The intensities of overtones diminish with increase of their frequencies. Each musical instrument produces different overtones and the overtones so present are normally a characteristic of those instruments. Even if two different musical instruments produce notes of same pitch, the overtones accompanying will be different. This difference defines the quality of the note emitted by one instrument and distinguishes the note produced by that instrument from that of the other. Waveforms of

the same note of fundamental frequency 440 Hz played on a violin and a piano are shown in Fig.36(a)

They are very different and provide visual evidence that the notes are of different 'quality'. Using a mathematical technique known as *Fourier analysis*, the frequencies and amplitudes of the harmonics present as overtones can be worked out in each case and sound spectra obtained (Fig.36(b)). Second and fifth harmonics are present with higher amplitude in the violin spectrum while their amplitudes are very small in the piano spectrum. Thus we find that *the quality of the note is marked by the nature of the composition of the frequencies of the note. It depends upon the presence or absence of particular overtones and their relative intensities.*

The quality plays a very important part in our day today life. Suppose a person is calling you. You can identify from the loudness and pitch whether it is a voice of a man or a woman or a child. The quality will help in identifying the particular person calling you. Nature has provided different overtones in the voices of different persons. Due to the *quality* of sound, we can recognise the person from his/her voice without seeing him/her.

9.10.8 Musical stone pillars

South Indian temples are famous for the stone pillars which emit musical notes when tapped. A single pillar is separated into columns of smaller, slender pillars. These pillars when stuck or tapped make different musical notes like an instrument. Many temples in south India such as Hampi in Karnataka, Meenakshi temple at Maduari, Vishnu temple at Alvar Tirunagar, Nellaiyapper temple in Tirunelveli and Thanumalaya temple at Suchindram have such musical stone pillars. Many temples were constructed during 16th century A.D. The sthapathis (sculptors) were master craftsmen with well versed knowledge in acoustics. With great difficulties they searched and selected resonant stones to carve such unique marvels. Fig.37 shows Vittala temple, Hampi containing rows of musical pillars carved in stones.



Fig. 37 Vittala temple in Hampi, Karnataka containing rows of musical pillars carved in stones

9.10.9 Limits of audibility

The limits of audibility of sound depend upon the intensity and frequency of sound. A sound is audible above a certain minimum intensity and a certain minimum frequency. These minimum limits are called *threshold intensity* of audibility and *lower pitch limit* of audibility respectively.

When the intensity of sound exceeds a maximum limit called *maximum audible intensity limit* it produces a sensation of pain on the ear. Similarly there is a max-

imum frequency limit called **upper pitch limit of audibility**, beyond which the sound is not audible. The minimum audible intensity and the maximum audible intensity vary with the frequency of sound. The limits of audibility can be understood well with the help of an audiogram given in Fig.38. In audiogram, along x -axis the frequency is plotted on a logarithmic scale and along y -axis the intensity (pressure in N/m^2) is plotted on a logarithmic scale. The two curves, one representing the threshold of audibility and another representing the threshold of feeling enclose an area called the **auditory sensation area**. If the pitch and the intensity of any sound are beyond the limits set by this area, the sound is not heard by human beings.

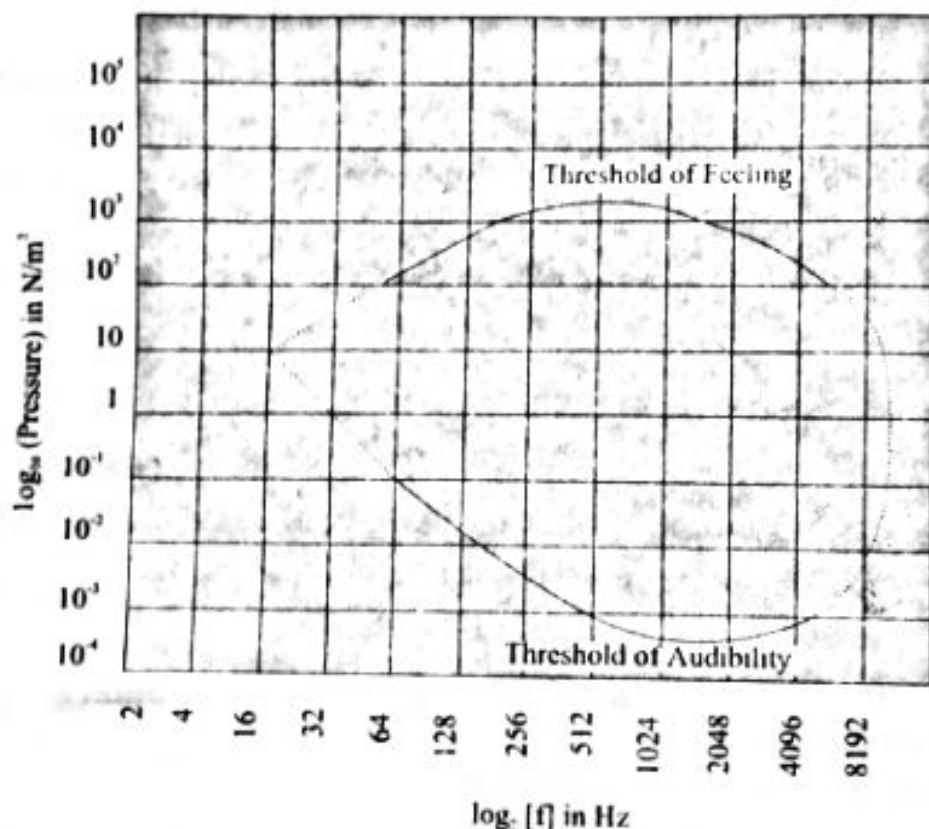


Fig.38 Audiogram

9.11 Sound Intensity in a Room

A room with hard, bare walls, no furniture and no drapes, *echoes*. To hear well in a room or auditorium it may seem that the walls should not absorb any sound. This would lead to echoing. A room, especially a large room with an excess of sound absorbing materials such as large soft drapes and soft and stuffed furniture, may have a quality referred to as *dead*. Too much absorption results in too low an intensity. Both these types of rooms are called **poor acoustics**. A balance between these two extremes must be therefore obtained.

9.12 Reverberation

When the sound source is switched on, the intensity slowly builds up; when it is switched off, the intensity drops slowly. *The prolongation of sound inside a room or hall even after the source producing the sound is turned off is called*

reverberation. This is due to multiple reflections from the walls, ceiling, floor and other reflecting materials present in the hall.

The reverberation time for a room is the time required for the intensity of sound to drop to one millionth (10^{-6}) of its initial value.

Reverberation time can be expressed in terms of sound level (in dB) rather than intensity. If the initial intensity is I_i , and the final intensity I_f is $10^{-6} I_i$, then

$$\text{Initial sound intensity level } dB_i = 10 \log I_i/I_o$$

$$\text{Final sound intensity level } dB_f = 10 \log I_f/I_o$$

$$dB_i - dB_f = 10 \log (I_i/I_f)$$

$$\text{Since } I_i/I_f = 10^6$$

$$dB_i - dB_f = 10 \log 10^6$$

$$= 6 \times 10 = 60 \quad (14)$$

The reverberation time is the time required for the intensity of sound to drop by 60 decibels.

It depends on the volume V of the room. It also depends on the absorption of all parts of the room, such as walls, furniture, people and so forth. Some parts may be highly absorbent, and some absorb only a little.

9.13 Absorption Coefficient

Since different materials absorb sound energy differently, absorption of all the materials are expressed in terms of absorption coefficient.

The coefficient of absorption of a material is defined as the ratio of the sound energy absorbed by the surface to that of the total sound energy incident on the surface.

$$\text{i.e., Absorption coefficient } a = \frac{\text{Sound energy absorbed by the surface}}{\text{Total sound energy incident on the surface}}$$

As all sound waves falling on an open window pass through, an open window is taken to be a perfect absorber of sound and absorption coefficient of all substances are measured in terms of open window unit (O.W.U.). **Absorption coefficient of a surface is also defined as the reciprocal of its area which absorbs the same sound energy as absorbed by unit area of an open window.**

9.14 Sabine's Formula (Growth and Decay of Sound)

Now let us study how sound grows and decays inside a room and hence derive an expression for reverberation time inside a room of volume V . Sound is produced by a source inside the room. These sound waves spread and fall on the walls; they are partly absorbed and partly reflected. The sound energy inside the room at any instant is given by

$$T = \frac{0.165 \text{ V}}{A} = \frac{0.165 \text{ V}}{\Sigma aS}$$

Solved Problem 22

The volume of a hall is 475 m^3 . The area of wall is 200 m^2 , area of floor and ceiling each of 100 m^2 . If absorption coefficient of the wall, ceiling and floor are 0.025, 0.02 and 0.55 respectively, calculate the reverberation time for the hall.

Solution

$$\text{Reverberation Time } T = \frac{0.165 V}{\Sigma aS}$$

$$\begin{aligned}\Sigma aS &= (200 \times 0.025 + 100 \times 0.02 + 100 \times 0.55) \\ &= 5 + 2 + 55 \\ &= 62\end{aligned}$$

$$\begin{aligned}\text{Therefore } T &= \frac{0.165 \times 475}{62} \\ &= 1.264 \text{ s}\end{aligned}$$

(Answer)

Solved Problem 23

A hall has a volume of 12500 m^3 and reverberation time of 1.5 sec. If 200 cushioned chairs are additionally placed in the hall, what will be the new reverberation time of the hall. The absorption of each chair is 1.0 O.W.U.

Solution

$$\text{Reverberation Time } T = \frac{0.165 V}{\Sigma aS}$$

Before adding the cushioned chairs

$$\text{reverberation time } T_1 = \frac{0.165 V}{\Sigma a S} \text{ i.e., } \Sigma a S = \frac{0.165 \times 12500}{1.5} \\ = 1375$$

After adding 200 cushioned chairs having absorption of 1 O.W.U. each

$$T_2 = \frac{0.165 V}{\Sigma a S + a_1 S_1} = \frac{0.165 \times 12500}{1375 + 200} \\ = 1.31 \text{ s} \quad (\text{Answer})$$

Solved Problem 24

A cinema hall has a volume of 7500 m^3 . What should be the total absorption in the hall if the reverberation time of 1.5 seconds is to be maintained?

Solution

$$\text{Reverberation Time } T = \frac{0.165 V}{\Sigma a S}$$

$$\text{Hence total absorption } \Sigma a S = \frac{0.165 V}{T} \\ = \frac{0.165 \times 7500}{1.5} \\ = 825 \text{ OWU} \quad (\text{Answer})$$

Solved Problem 25

A hall of volume $4 \times 6 \times 10 \text{ m}^3$ has a reverberation time 1.5s.

- (a) What is the total sound absorption of the hall?
- (b) 40 visitors enter the hall. Find the new reverberation time if each person is equivalent to 0.5 metric Sabine sound absorption.

Solution

$$(a) \text{ Formula } T = \frac{0.165 V}{\Sigma a S}$$

$$\text{Total sound absorption } \Sigma a S = \frac{0.165 V}{T} \\ = \frac{0.165 \times 4 \times 6 \times 10}{1.5} \\ = 26.4 \text{ metric Sabine or OWU}$$

(b) After 40 visitors enter the total sound absorption

$$\begin{aligned} \text{changes to} &= 26.4 + (40 \times 0.5) \\ &= 46.4 \text{ metric Sabine or OWU} \end{aligned}$$

$$\begin{aligned} \text{Hence new reverberation time } T &= \frac{0.165 \times 4 \times 6 \times 10}{46.4} \\ &= 0.8535 \text{ s} \end{aligned}$$

(Answer)

Solved Problem 26

A rectangular closed room whose internal dimensions are $20 \times 30 \times 16 \text{ m}^3$ has a surface whose average absorption coefficient is 0.04. Calculate the time of reverberation of the room. When a material of area 2000 m^2 is brought into the room, the time of reverberation falls to 3s. Calculate the absorption coefficient of the material introduced.

Solution

$$T = \frac{0.165V}{\Sigma aS}$$

$$\text{Volume of the room } V = 20 \times 30 \times 16 = 9600 \text{ m}^3$$

$$\text{Average absorption coefficient } a = 0.04$$

$$\text{Surface area } S = \text{area of the floor + ceiling + walls}$$

$$\begin{aligned} &= (20 \times 30) + (20 \times 30) + 2(20 \times 16) + 2(30 \times 16) \\ &= (1200 + 640 + 960) = 2800 \text{ m}^2 \end{aligned}$$

$$\text{Hence reverberation time } T = \frac{0.165 \times 9600}{0.04 \times 2800}$$

$$= 14.143 \text{ s} \quad (\text{Answer})$$

After introducing a material of area 2000 m^2 , T falls to 3s.

$$T = \frac{0.165V}{\Sigma aS + a_1s_1}$$

$$\Sigma aS + a_1s_1 = \frac{0.165V}{T}$$

$$a_1s_1 = \frac{0.165V}{T} - \Sigma aS$$

$$\text{or } a_1 = \frac{0.165V}{S_1T} - \frac{\Sigma aS}{S_1}$$

$$= \frac{0.165 \times 9600}{2000 \times 3} - \frac{2800 \times 0.04}{2000}$$

$$= 0.264 - 0.056$$

(Answer)

i.e. absorption coefficient of the material introduced = 0.208